SAILESH VEMULA

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PROFESSIONAL SUMMARY

Senior Data Engineer with 6+ years of experience designing and optimizing large-scale, high-availability data platforms across AWS and Azure cloud environments. Expert in building real-time streaming pipelines and ETL/ELT frameworks using PySpark, SQL, Flink, AWS Glue, and Redshift, handling multi-terabyte daily workloads. Proven track record in data modeling, Iceberg/Delta Lake management, and query optimization, implementing cost-efficient architectures that reduce processing latency from hours to minutes. Adept at modern DataOps, CI/CD, and cloud-native best practices to deliver governed, secure, and scalable data ecosystems enabling predictive analytics, machine learning, and BI reporting.

TECHNICAL SKILLS

Programming Languages : Python (Pandas, NumPy, PySpark, Scrapy), SQL, PL/SQL, Scala, TypeScript, Bash

Databases & Warehousing: Snowflake, Redshift, DynamoDB, PostgreSQL, SQL Server, MySQL, MongoDB, StarRocks, Athena, RDS

Big Data & Streaming : Apache Spark, Iceberg , Hadoop, Apache Kafka, Apache Flink, Kinesis, Amazon MSK

Cloud & Orchestration: AWS (S3, Glue , EMR, Lambda, Lake Formation, EventBridge, CloudTrail, KMS, SageMaker, EC2, EKS), Azure Data Lake, Terraform, Docker, Airflow, Step Functions, SSIS

Visualization & Monitoring : QuickSight, Tableau, Power BI, SSRS, Prometheus, Cloudwatch

DataOps and Methodologies : CI/CD (GitHub, Git, AWS CodePipeline), Git, Agile, SDLC

EXPERIENCE

Amazon — Mar 2022 - Present
Data Engineer II — **Austin, TX**

- Automated roster audits on **10M+ records daily** by integrating **Airflow**, **DynamoDB**, **Redshift**, **SNS**, **SQS** and APIs. Generated multilingual alerts that reduced compliance resolution time by **60%** and served as a real-world telemetry system for global operations.
- A **1TB/day ETL** framework using **AWS Glue, PySpark,** and **S3** brought data from RDS, Redshift, Lake formation tables and DynamoDB into **S3 Data Lake**, powering a metrics data platform for 10K+ monthly users and **removing 15,000+ hours of manual** reporting work each month.
- Introduced an attendance anomaly detection service with **Python** and **Airflow DAGs**, which automatically raised alerts (Emails, Calls, Messages), saving **5,260+ manual hours** yearly and enabling HR to respond faster to staffing issues.
- Designed pipelines leveraging AWS Glue, **PySpark**, and S3 to prepare knowledge bases for training context-based RAG models, enabling integration of diverse datasets and powering a final **RAG** system that could accurately answer user questions about enterprise reports.
- Reliability improved when **EventBridge, Lambda, and CloudWatch** were used to validate incoming events; this raised overall data quality from **68% to 98%**, eliminating errors that previously impacted downstream systems. Build **Quicksight** dashboards that can be shared across the leadership for tracking the efficiency.
- ETL throughput scaled further after optimizing queries, performance tuning, PySpark **partitioning**, **job parallelism**, and **caching** in AWS Glue, Redshift which cut large-volume job runtimes by **30%** and increased the speed of workforce analytics delivery.

Root Insurance — Jun 2021 - Mar 2022
Data Engineer — **Austin, TX**

- Delivered a distributed AWS **Redshift** data warehouse with dimensional modeling, consolidating policy, claims, and telematics data into a single analytics source that accelerated enterprise reporting cycles.
- Automated ingestion of **6TB+ daily** insurance data by orchestrating AWS Glue, Step Functions, and S3, enabling machine learning pipelines to update risk-based pricing in near real time.
- Integrated APIs, SQL Server, and CSV feed into Redshift and **Snowflake** while embedding Python- and SQL-driven data validation, maintaining **99.9% dataset accuracy** for downstream analytics.
- Reduced monthly infrastructure costs by **$6K** through migrating six high-volume workloads from SQL Server,

DynamoDB, and S3 into Redshift with optimizing distribution strategies.

- Equipped executives with **20+ Tableau dashboards** that refreshed hourly from Redshift, providing timely insight into underwriting risk trends and claims performance metrics.
- Increased Redshift query **throughput by 55%** via schema redesign, sort/partition key adjustments, and Workload Management tuning, improving the responsiveness of operational dashboards.

Indiana Business Research Center Jan 2020 - Dec 2020
Data Engineer - Intern **Indianapolis, IN**

- Developed a Microsoft **SQL Server Master Data Management** system with automated **SSIS** and **Python ETL** jobs, unifying pandemic-related data from multiple international health agencies for centralized reporting.
- **Migrated terabytes** of healthcare records into an Azure Data Lake with optimized partitioning, cutting query times by **40%** while ensuring HIPAA compliance with sensitive patient information.

Tata Consultancy Services Sep 2017 - Jul 2019
Data Engineer **Hyderabad, India**

- Automated data ingestion processes by building Python-based SSIS jobs, which reduced a **2+ hour** manual workloads to minutes and ensured on-time availability of operational datasets in the SQL Server and Power BI Dashboards.
- Cut SQL Server query **execution times by 60%** through careful tuning of indexes, stored procedures, and partition strategies, enabling faster data retrieval for analytics users.
- Consolidated multiple data sources, including **flat files, APIs**, and relational systems, into a centralized SQL Server data mart, streamlining reporting across diverse business units.
- Introduced reusable Python modules for validation and transformation, improving dataset reliability by 25%, and reducing reprocessing caused by data quality issues.

PROJECTS

Data Visualization of US Foreign Trade Statistics

- Ingested and standardized US Census trade datasets across all states, ensuring consistency in commodity codes, units, and valuation metrics for downstream analytics and economic research.
- Built interactive dashboards using Python and Pandas to visualize trade volumes, top commodities, and value trends across geographies, implementing drilldowns and time-series analysis that accelerated stakeholder insights.

Predictive Analysis on Zomato Restaurant Data Using Machine Learning

- Processed 1GB+ of restaurant data using Spark RDDs and Spark SQL, applying feature engineering to derive predictors for ratings and costs, and developed TensorFlow models with hyperparameter tuning for improved rating prediction accuracy.
- Applied NLP sentiment analysis to customer reviews using GPU-accelerated training on SageMaker, reducing model runtime and delivering actionable insights for business optimization.

Dashboard for World Population Statistics

- Consolidated global demographic datasets from multiple open data sources into a structured SQL data warehouse and developed an interactive Tableau dashboard displaying population growth, density, and demographic distribution trends across continents and countries.
- Incorporated time-series analytics and calculated KPIs to track population changes over decades, enabling comparative regional insights and data-driven demographic analysis.

EDUCATION

MS in Applied Data Science **Dec 2020**
Indiana University Bloomington
Bachelor of Technology in Electronics and Communication **May 2017**
V R Siddhartha Engineering College | Vijayawada, India